The Learning Premium in the Large-Horizon Limit

Propositions 1–2 (draft for *Orchestrating Agent Teams*)

Working draft

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Abstract

We show that, as the campaign horizon K grows with the budget per period $\rho = B/K$ held fixed, the per-period learning premium of the learn-then-exploit (ETC) team converges to an explicit function of the primitives: the value of the best binomial probing experiment, which coincides with the value of free perfect information $\mathrm{VPI}(\rho)$ outside a neighbourhood of the entry threshold. The limit does not depend on how the team schedules its learning, so the direction of the planner's reallocation in the limit is pinned down by primitives alone (Proposition 1 and Corollary 1). We then characterise the shape of the limit premium (Proposition 2): it is negative-to-zero at small budgets (information, even if free, lowers the value of a budget-paced team), positive and decaying like $\mathrm{Var}(\theta)/\rho^2$ at large budgets, and it opens with a kink, so the marginal-value ratio jumps at entry. The upper edge of the learning window is a finite-horizon effect.

1 Setup and notation

We keep the model of the paper. The market state is $\theta \in \{\theta_L, \theta_H\}$ with $0 < \theta_L < \theta_H$ and prior $\pi_0 = \Pr(\theta = \theta_H) \in (0,1)$; $r = \theta_H/\theta_L$, $b^I = \sqrt{\theta_L\theta_H}$ and $\kappa_0 = \pi_0/(1-\pi_0)$. A bid b wins with probability $q_\theta(b) = b/(b+\theta)$ and pays b upon winning; $\bar{q}(\pi, b) = \pi q_{\theta_H}(b) + (1-\pi)q_{\theta_L}(b)$. A team has budget B and horizon K; we write $\rho = B/K$ for the budget per period and index values by K when the horizon matters.

Budget accounting and the rate function. As in the paper, whenever a team commits to an exploitation plan at belief π with remaining budget R over n periods, the plan spends R in expectation under π. By Lemma 2 of the paper (Exploitation phase) the optimal plan is a constant paced bid and its value is $n\, g(\pi; R/n)$, where for $\pi \in [0,1]$ and $\rho \geq 0$

$$g(\pi;\rho) := \bar{q}\big(\pi, b^*(\pi,\rho)\big), \qquad b^*(\pi,\rho)\,\bar{q}\big(\pi, b^*(\pi,\rho)\big) = \rho. \tag{1}$$

In particular the no-learning benchmark is $W_K(B) = K\, g(\pi_0; B/K)$.

Learn-then-commit protocols. A protocol in the class $\mathcal{L}$ probes at a fixed bid $b_p > 0$ for σ periods, where $\sigma \leq K-1$ is a stopping time of the probe outcomes, and hands the number of wins w_σ to an exploitation agent that commits at the posterior π_σ to the paced plan with the remaining budget $R_\sigma = B - w_\sigma b_p \geq 0$ over the remaining $K-\sigma$ periods. By Lemma 2 of the paper its value is

$$V_K^\sigma(B) = \mathbb{E}\Big[w_\sigma + (K-\sigma)\, g\Big(\pi_\sigma; \frac{R_\sigma}{K-\sigma}\Big)\Big]. \tag{2}$$

The paper's ETC is the case $b_p = b^I$ and $\sigma \equiv \tau$ deterministic, with $V_K(B) = \max\{V_K^\tau(B) : 0 \le \tau \le \min(K-1, B/b^I)\}$; the adaptive-stopping version (draft DP) is $b_p = b^I$ with σ chosen optimally. Expectations are under the prior-predictive law, under which the posterior $(\pi_t)_t$ is a martingale with $\mathbb{E}[\pi_t] = \pi_0$.

Definition 1 (Limit objects)**.** For $\rho > 0$ let π_τ denote the posterior after τ probes at b^I and $\pi_\infty := \mathbf{1}_{\{\theta=\theta_H\}}$. Define

$$\begin{aligned}
\mathrm{VPI}(\rho) &:= \pi_0\, g(1;\rho) + (1-\pi_0)\, g(0;\rho) - g(\pi_0;\rho),\\
f_\tau(\rho) &:= \mathbb{E}\big[g(\pi_\tau;\rho)\big] - g(\pi_0;\rho) \quad (\tau \in \mathbb{N}), \qquad f_\infty := \mathrm{VPI},\\
\Psi(\rho) &:= \max\Big\{0,\ \sup_{\tau\in\mathbb{N}\cup\{\infty\}} f_\tau(\rho)\Big\},\\
\Phi(\rho) &:= \hat g(\pi_0;\rho) - g(\pi_0;\rho),
\end{aligned}$$

where $\hat g(\cdot;\rho)$ is the concave envelope of $g(\cdot;\rho)$ on $[0,1]$. Thus VPI is the per-period value of free perfect information, Ψ the value of the best binomial probing experiment at b^I (including none), and Φ the value of the best information structure. Since $\mathbb{E}[\pi_\tau] = \pi_0$ and $\hat g$ is concave, $\max\{\mathrm{VPI}(\rho), 0\} \le \Psi(\rho) \le \Phi(\rho)$.

Lemma 1 (Properties of g)**.** *(i) For every π, $\rho \mapsto g(\pi;\rho)$ is continuous, strictly increasing and strictly concave on $[0,\infty)$, with $g(\pi;0) = 0$ and $g < 1$.*

(ii) For every π and $S \ge 0$, $n \mapsto n\, g(\pi; S/n)$ is nondecreasing on $(0,\infty)$.

(iii) For $\rho > 0$, $g(\cdot;\rho) \in C^\infty([0,1])$, and g, $\partial_\pi g$, $\partial_\rho g$ are bounded on $[0,1]\times[\rho_a,\rho_b]$ for every $0 < \rho_a < \rho_b < \infty$.

Proof. (i) is shown in the proof of Lemma 2 of the paper. (ii) Concavity and $g(\pi;0) = 0$ imply that $x \mapsto g(\pi;x)/x$ is nonincreasing on $(0,\infty)$; hence $n\, g(\pi; S/n) = S \cdot g(\pi;S/n)/(S/n)$ is nondecreasing in n. (iii) $F(b,\pi,\rho) = b\,\bar q(\pi,b) - \rho$ is rational in (b,π) with positive denominators and $\partial_b F = \bar q + b\,\partial_b \bar q > 0$ for $b > 0$; the implicit function theorem gives a C^∞ solution $b^*(\pi,\rho)$ for $\rho > 0$, and boundedness follows from continuity on compact sets. □

Lemma 2 (Posterior concentration at b^I)**.** *Let $\beta := 2r^{1/4}/(1+\sqrt{r}) \in (0,1)$, the Bhattacharyya coefficient of the two outcome distributions at b^I. For every $\tau \in \mathbb{N}$, $\mathbb{E}\big[|\pi_\tau - \pi_\infty|\big] \le 2\sqrt{\pi_0(1-\pi_0)}\,\beta^\tau$.*

Proof. At b^I the win probabilities are $p_H = 1/(1+\sqrt{r})$ and $p_L = \sqrt{r}/(1+\sqrt{r})$, and the posterior odds are $\kappa_\tau = \kappa_0 r^{(\tau-2w)/2}$. Under θ_H, $1-\pi_\tau = 1/(1+\kappa_\tau) \le \kappa_\tau^{-1/2}$ and each probe multiplies $\kappa^{-1/2}$ by $r^{1/4}$ (win) or $r^{-1/4}$ (loss), so $\mathbb{E}[1-\pi_\tau \mid \theta_H] \le \kappa_0^{-1/2}\big(p_H r^{1/4} + (1-p_H) r^{-1/4}\big)^\tau = \kappa_0^{-1/2}\beta^\tau$. Symmetrically $\mathbb{E}[\pi_\tau \mid \theta_L] \le \kappa_0^{1/2}\beta^\tau$. Averaging with weights $\pi_0, 1-\pi_0$ gives the bound. Finally $\beta < 1$ because $1+\sqrt{r} > 2r^{1/4}$ for $r \ne 1$. □

2 Proposition 1: the large-horizon limit

Proposition 1 (Large-horizon limit of the learning premium)**.** *Fix $\rho > 0$ and let $B = \rho K$.*

(a) Achievability. *There are constants c_ρ, c'_ρ (depending on the primitives and ρ only) such that for every $\tau \in \mathbb{N}$ and all K large enough*

$$\frac{V_K^\tau(\rho K) - W_K(\rho K)}{K} \;\ge\; f_\tau(\rho) - c_\rho \frac{\tau}{K},$$

and for $\tau_K = \lceil \log K / \log(1/\beta) \rceil$,

$$\frac{V_K^{\tau_K}(\rho K) - W_K(\rho K)}{K} \;\geq\; \mathrm{VPI}(\rho) - c'_\rho \, \frac{1 + \log K}{K}.$$

Consequently $\liminf_K [V_K(\rho K) - W_K(\rho K)]/K \geq \Psi(\rho)$.

(b) Upper bound. *For every protocol in* $\mathcal{L}$ *and every* K, B,

$$\frac{V_K^{\sigma}(B) - W_K(B)}{K} \;\leq\; \frac{\mathbb{E}[\sigma]}{K} + \Phi(B/K),$$

and for the paper's ETC with a deterministic τ, $[V_K^{\tau}(B) - W_K(B)]/K \leq \tau/K + \Psi(B/K)$.

(c) Limit. *The optimal probing length satisfies* $\tau_K^*/K \to 0$, *and*

$$\lim_{K \to \infty} \frac{V_K(\rho K) - W_K(\rho K)}{K} = \Psi(\rho).$$

(d) Protocol invariance. *For every sequence of protocols in* $\mathcal{L}$ *with* $\mathbb{E}[\sigma_K] = o(K)$,

$$\limsup_K \; [V_K^{\sigma_K}(\rho K) - W_K(\rho K)]/K \;\leq\; \Phi(\rho).$$

On the set $\mathcal{R} := \{\rho : \Phi(\rho) = \max\{\mathrm{VPI}(\rho), 0\}\}$ *the paper's ETC is asymptotically optimal in this class, and its limit premium* $\max\{\mathrm{VPI}(\rho), 0\}$ *depends on the primitives* $(\theta_L, \theta_H, \pi_0)$ *only.*

Proof. *(a)* Fix τ; for K large, $\tau \leq K-1$ and $\tau b^I \leq \rho K$, so τ is feasible. Since $w \leq \tau$, the exploitation budget per period satisfies $R/(K-\tau) \geq \rho_K^\tau := \rho - \tau b^I/(K-\tau) \geq \rho/2$. Dropping the probe wins in (2) and using monotonicity and the Lipschitz bound $L_\rho := \sup\{\partial_\rho g(\pi; x) : \pi \in [0,1], x \in [\rho/2, \rho]\}$ from Lemma 1(iii),

$$V_K^\tau(\rho K) \geq (K-\tau)\,\mathbb{E}\big[g(\pi_\tau; \rho_K^\tau)\big] \geq (K-\tau)\,\mathbb{E}\big[g(\pi_\tau; \rho)\big] - L_\rho \tau b^I \geq K\,\mathbb{E}\big[g(\pi_\tau; \rho)\big] - \tau(1 + L_\rho b^I),$$

using $g < 1$. Subtracting $W_K(\rho K) = K g(\pi_0; \rho)$ gives the first bound with $c_\rho = 1 + L_\rho b^I$. For $\tau = \tau_K$, the mean value theorem with $M_\rho := \max_\pi |\partial_\pi g(\pi; \rho)|$ and Lemma 2 give $f_{\tau_K}(\rho) \geq \mathrm{VPI}(\rho) - 2M_\rho \sqrt{\pi_0(1-\pi_0)}\,\beta^{\tau_K} \geq \mathrm{VPI}(\rho) - M_\rho/K$, since $\beta^{\tau_K} \leq 1/K$; combining with the first bound and $\tau_K \leq 1 + \log K/\log(1/\beta)$ proves the second. Since $V_K \geq V_K^\tau$ for every feasible τ, $\liminf_K [V_K - W_K]/K \geq f_\tau(\rho)$ for all $\tau \in \mathbb{N} \cup \{\infty\}$, and $\tau = 0$ gives 0; hence the lim inf is at least $\Psi(\rho)$.

(b) In (2), $R_\sigma \leq B$, monotonicity and Lemma 1(ii) give $(K-\sigma)\, g(\pi_\sigma; R_\sigma/(K-\sigma)) \leq (K-\sigma)\, g(\pi_\sigma; B/(K-\sigma)) \leq K\, g(\pi_\sigma; B/K)$, and $w_\sigma \leq \sigma$. By optional stopping $\mathbb{E}[\pi_\sigma] = \pi_0$, so by concavity of $\hat{g}$, $\mathbb{E}[g(\pi_\sigma; \rho)] \leq \mathbb{E}[\hat{g}(\pi_\sigma; \rho)] \leq \hat{g}(\pi_0; \rho)$. Hence $V_K^\sigma(B) \leq \mathbb{E}[\sigma] + K\hat{g}(\pi_0; \rho)$; subtract $W_K(B) = K g(\pi_0; \rho)$. For deterministic τ at b^I, replace $\hat{g}(\pi_0; \rho)$ by $\mathbb{E}[g(\pi_\tau; \rho)] = g(\pi_0; \rho) + f_\tau(\rho) \leq g(\pi_0; \rho) + \Psi(\rho)$.

(c) Suppose $\limsup_K \tau_K^*/K > 0$ and pass to a subsequence with $\tau_K^*/K \to \varepsilon \in (0, 1]$. Feasibility ($\tau b^I \leq \rho K$) gives $\varepsilon b^I \leq \rho$. Since $\tau_K^* \to \infty$, conditional on θ the strong law gives $w/\tau_K^* \to q_\theta(b^I)$ and Lemma 2 gives $\pi_{\tau_K^*} \to \pi_\infty$. With $s_\theta := b^I q_\theta(b^I)$, dominated convergence in (2) yields

$$\lim_K \frac{V_K^{\tau_K^*}(\rho K)}{K} = \mathbb{E}_\theta\Big[\varepsilon\, q_\theta(b^I) + (1-\varepsilon)\, g\Big(\pi_\infty; \tfrac{\rho - \varepsilon s_\theta}{1-\varepsilon}\Big)\Big]$$

(for $\varepsilon = 1$ the second term is absent). Because b^I is the paced bid for spend s_θ in a known state, $q_\theta(b^I) = g(\pi_\infty; s_\theta)$ on $\{\theta\}$, and strict concavity of $g(\pi_\infty; \cdot)$ gives $\varepsilon g(\pi_\infty; s_\theta) + (1-\varepsilon) g(\pi_\infty; \frac{\rho - \varepsilon s_\theta}{1-\varepsilon}) \leq$

$g(\pi_\infty;\rho)$ with equality only if $s_\theta = \rho$ (for $\varepsilon = 1$: $g(\pi_\infty;s_\theta) \le g(\pi_\infty;\rho)$ because $s_\theta < b^I \le \rho$). As $s_{\theta_L} \neq s_{\theta_H}$, the inequality is strict for at least one state, so $\lim_K V_K^{\tau_K^*}/K < \mathbb{E}_\theta[g(\pi_\infty;\rho)] = g(\pi_0;\rho) + \mathrm{VPI}(\rho) \le g(\pi_0;\rho) + \Psi(\rho)$, which by (a) is at most $\liminf_K V_K/K$ — contradicting $V_K = V_K^{\tau_K^*}$. Hence $\tau_K^* = o(K)$, and (b) gives $\limsup_K [V_K - W_K]/K \le \Psi(\rho)$; together with (a) the limit is $\Psi(\rho)$.

(d) The first claim is (b). On $\mathcal{R}$, $\max\{\mathrm{VPI},0\} \le \Psi \le \Phi = \max\{\mathrm{VPI},0\}$, so the ETC limit in (c) attains the upper bound. □

Corollary 1 (Allocation in the limit)**.** *Let $L(\rho) := g(\pi_0;\rho)+\Psi(\rho)$ for $\rho > 0$ and $L(0) = 0$. Fix $v_1 > v_2 > 0$ and $t > 0$, and let $T = tK$. Then $a_K^E/K = \alpha^E(t) := \arg\max_{\alpha\in[0,t]}\{v_1 g(\pi_0;\alpha)+v_2 g(\pi_0;t-\alpha)\}$ for every K, and every limit point of a_K^L/K maximizes $v_1 L(\alpha) + v_2 L(t-\alpha)$ over $\alpha \in [0,t]$. If that maximizer $\alpha^L(t)$ is unique, $(a_K^L - a_K^E)/K \to \alpha^L(t) - \alpha^E(t)$. At differentiability points and under the concavity condition of Theorem 1 of the paper, the sign of the limit shift is the sign of*

$$\frac{L'(\alpha^E)}{g_\rho(\pi_0;\alpha^E)} - \frac{L'(t-\alpha^E)}{g_\rho(\pi_0;t-\alpha^E)},$$

so the direction of the reallocation in the limit is determined by the primitives alone.

Proof. $W_K(\alpha K) = K g(\pi_0;\alpha)$ makes the benchmark problem independent of K. For each K, $\rho \mapsto V_K(\rho K)/K$ is nondecreasing (a larger budget enlarges the feasible set of τ and raises every exploitation value) and converges pointwise to L by Proposition 1(c). The family $\{f_\tau\}$ is equi-Lipschitz on compact subsets of $(0,\infty)$ (Lemma 1(iii)), so Ψ is continuous there, and $\Psi = 0$ near 0 by Proposition 2(a); thus L is continuous on $[0,\infty)$. Pointwise convergence of monotone functions to a continuous limit on a compact interval is uniform (Pólya), so the planner's normalised objectives converge uniformly on $[0,t]$ and limit points of maximisers maximise the limit objective. The sign statement is Theorem 1 of the paper applied to L, noting $W_K'(B) = g_\rho(\pi_0;B/K)$. □

Remark 1 (Learning while exploiting). Proposition 1 covers every learn-then-commit protocol. A team that updates after every auction and re-paces its remaining budget (certainty-equivalent pacing, "CE") is not in $\mathcal{L}$. In this model every bid is informative ($q_{\theta_L}(b) \neq q_{\theta_H}(b)$ for all $b > 0$), so such a team learns the state while exploiting, and the numerical evidence in Section 4 shows that its per-period premium converges to $\mathrm{VPI}(\rho)$ with a learning cost that grows much more slowly in K than that of fixed-τ ETC. A proof along the lines of (a) requires posterior concentration under bids bounded away from 0 and ∞ and control of the realised-wallet re-pacing; we leave it for future work.

3 Proposition 2: the shape of the limit premium

Let $m_k(\pi) := \pi\theta_H^{-k} + (1-\pi)\theta_L^{-k}$ and $\mu_k(\pi) := \pi\theta_H^k + (1-\pi)\theta_L^k$ (the k-th moments of $1/\theta$ and θ under belief π), and let $\mathrm{Var}_{\pi_0}(\theta) = \pi_0(1-\pi_0)(\theta_H - \theta_L)^2$.

Proposition 2 (Shape of the limit premium)**.** *(a)* Small budgets. *As $\rho \downarrow 0$, uniformly in $\pi \in [0,1]$ together with the first two π-derivatives,*

$$g(\pi;\rho) = \sqrt{\rho\, m_1(\pi)} + O(\rho).$$

Hence $\mathrm{VPI}(\rho) = -c_1\sqrt{\rho} + O(\rho)$ with $c_1 := \sqrt{m_1(\pi_0)} - \pi_0\theta_H^{-1/2} - (1-\pi_0)\theta_L^{-1/2} > 0$, and there is $\rho_c > 0$ such that $g(\cdot;\rho)$ is strictly concave on $[0,1]$ for all $\rho < \rho_c$; consequently $\Phi(\rho) = \Psi(\rho) = 0$ there: no information structure, free or costly, raises the value of a budget-paced team.

(b) Large budgets. *As $\rho \to \infty$, uniformly in $\pi \in [0,1]$ together with the first two π-derivatives,*

$$g(\pi;\rho) = 1 - \frac{\mu_1(\pi)}{\rho} + \frac{\mu_1(\pi)^2 + \mu_2(\pi)}{\rho^2} + O(\rho^{-3}).$$

Hence $\mathrm{VPI}(\rho) = \mathrm{Var}_{\pi_0}(\theta)/\rho^2 + O(\rho^{-3})$*, and there is $\rho_v < \infty$ such that $g(\cdot;\rho)$ is strictly convex on $[0,1]$ for all $\rho > \rho_v$; consequently $\Psi(\rho) = \Phi(\rho) = \mathrm{VPI}(\rho) > 0$ there.*

(c) The learning window in the limit. *Let $\rho_1 := \inf\{\rho > 0 : \Psi(\rho) > 0\}$. Then $0 < \rho_c \le \rho_1 \le \rho_v < \infty$: learning has no value below ρ_1 and positive value at every $\rho > \rho_v$, decaying like $\mathrm{Var}_{\pi_0}(\theta)/\rho^2$. In the limit the window has no upper edge; in particular the finite-horizon upper edge $B_2(K)$ of the paper satisfies $B_2(K)/K \to \infty$.*

(d) Kink at entry. *If near ρ_1 one has $\Psi = \max\{0, f_\tau\}$ for some $\tau \in \mathbb{N} \cup \{\infty\}$ with $f_\tau(\rho_1) = 0 < f'_\tau(\rho_1)$, then L has a convex kink at ρ_1 and the limit marginal-value ratio $L'/g_\rho(\pi_0;\cdot)$ jumps from 1 to $1 + f'_\tau(\rho_1)/g_\rho(\pi_0;\rho_1)$.*

Proof. *(a)* Put $s = \sqrt{\rho}$ and $b = s\,x$. For $b < \theta_L$, $q_\theta(b) = \sum_{k\ge1}(-1)^{k-1}(b/\theta)^k$, so $b\,\bar q(\pi,b) = \sum_{k\ge1}(-1)^{k-1}m_k(\pi)\,b^{k+1}$ and the pacing equation $b\,\bar q = s^2$ reads

$$G(x,\pi,s) := \sum_{k\ge1}(-1)^{k-1}m_k(\pi)\,s^{k-1}x^{k+1} - 1 = 0.$$

G is real-analytic near $s = 0$, $G(x_0,\pi,0) = 0$ for $x_0(\pi) = m_1(\pi)^{-1/2}$, and $\partial_x G(x_0,\pi,0) = 2\sqrt{m_1(\pi)} > 0$. By the analytic implicit function theorem and compactness of $[0,1]$ there are $s_0 > 0$ and an analytic solution $x(\pi,s)$ on a neighbourhood of $[0,1]\times[0,s_0]$. Then $g(\pi;\rho) = \bar q(\pi, s\,x) = s\,h(\pi,s)$ with h analytic and $h(\pi,0) = m_1(\pi)x_0(\pi) = \sqrt{m_1(\pi)}$, which gives the expansion with all π-derivatives. Substituting $m_1(1) = \theta_H^{-1}$, $m_1(0) = \theta_L^{-1}$ yields the expansion of VPI; $c_1 > 0$ is Jensen's inequality for the strictly concave square root applied to the non-degenerate random variable $1/\theta$. Finally $\partial_\pi^2 g = -\sqrt{\rho}\,(m_1')^2/(4m_1^{3/2}) + O(\rho)$ with $m_1' = \theta_H^{-1} - \theta_L^{-1} \ne 0$ and $m_1 \in [\theta_H^{-1}, \theta_L^{-1}]$, so $g(\cdot;\rho)$ is strictly concave on $[0,1]$ for small ρ. Then $\hat g = g$, so $\Phi = 0$, and $f_\tau \le 0$ for every τ by Jensen, so $\Psi = 0$.

(b) Put $u = 1/\rho$ and $b = y/u$. Since $q_\theta(b) = 1 - u\theta/(y+u\theta)$, the pacing equation reads $H(y,\pi,u) := y - y\,\mathbb{E}_\pi[u\theta/(y+u\theta)] - 1 = 0$, where $\mathbb{E}_\pi$ averages over θ with $\Pr(\theta_H) = \pi$. H is analytic near $u = 0$, $H(1,\pi,0) = 0$ and $\partial_y H(1,\pi,0) = 1$, so $y(\pi,u)$ is analytic near $[0,1]\times\{0\}$ and so is $g = 1 - \mathbb{E}_\pi[u\theta/(y+u\theta)]$. Expanding, $y = 1 + u\mu_1 - u^2\mu_2 + O(u^3)$ and $g = 1 - u\mu_1 + u^2(\mu_1^2+\mu_2) + O(u^3)$. For a known state ($\pi \in \{0,1\}$) this is $1 - u\theta + 2u^2\theta^2 + O(u^3)$, and averaging over the prior, $\mathrm{VPI} = u^2\big(2\mu_2 - \mu_1^2 - \mu_2\big) + O(u^3) = u^2\,\mathrm{Var}_{\pi_0}(\theta) + O(u^3)$, where $\mu_k = \mu_k(\pi_0)$. Since μ_1, μ_2 are affine in π, $\partial_\pi^2 g = 2u^2(\theta_H - \theta_L)^2 + O(u^3) > 0$ for small u, so $g(\cdot;\rho)$ is strictly convex. Then $\hat g$ is the chord, so $\Phi = \mathrm{VPI}$; and because $\pi_{\tau+1}$ is a mean-preserving spread of π_τ, $\mathbb{E}[g(\pi_\tau;\rho)]$ is nondecreasing in τ with limit $\pi_0 g(1;\rho) + (1-\pi_0)g(0;\rho)$, so $\sup_\tau f_\tau = \mathrm{VPI}$ and $\Psi = \mathrm{VPI}$. Strict convexity also gives $\mathrm{VPI} > 0$.

(c) By (a), $\Psi = 0$ on $(0,\rho_c)$, so $\rho_1 \ge \rho_c > 0$; by (b), $\Psi > 0$ on (ρ_v,∞), so $\rho_1 \le \rho_v$. For any fixed $\rho > \rho_v$, Proposition 1(a) gives $[V_K(\rho K) - W_K(\rho K)]/K \ge \mathrm{VPI}(\rho) - c'_\rho(1+\log K)/K > 0$ for K large, so the ETC team learns at budget ρK and $B_2(K) \ge \rho K$ eventually; as ρ is arbitrary, $B_2(K)/K \to \infty$.

(d) Near ρ_1, $L = g(\pi_0;\cdot) + \max\{0, f_\tau\}$. Its left derivative at ρ_1 is $g_\rho(\pi_0;\rho_1)$ and its right derivative is $g_\rho(\pi_0;\rho_1) + f'_\tau(\rho_1)$, a convex kink. Dividing by $g_\rho(\pi_0;\rho_1)$ gives the jump of the ratio. □

Remark 2 (Interpretation). Part (a) is the limit form of the cross-state subsidy: at small budgets the uninformed paced bid spends more in the cheap state, where the marginal dollar buys more impressions, and any information that splits the team's belief removes this subsidy because each posterior must balance its own budget. Under an ex-ante budget constraint, by contrast, free information is never harmful (Blackwell). Part (b) is decision insensitivity: at large budgets both states are won almost surely and the value of information decays like $\mathrm{Var}(\theta)/\rho^2$, but it never vanishes; the upper edge of the paper's learning window comes from the cost $O(\log K)$ of probing at b^I, which is spread over the horizon and vanishes per period as $K \to \infty$. Together with Corollary 1, the two allocation regimes of the paper follow in the limit: if the junior team's budget per period is below ρ_1 and the senior's is where the limit ratio exceeds one, learning skews the split toward the senior team; once both are above ρ_1 and the limit ratio is decreasing between them, learning balances the split.

4 Numerical verification

All computations use the paper's baseline $\theta_L = \frac{1}{2}$, $\theta_H = 8$ ($r = 16$, $b^I = 2$), $\pi_0 = \frac{1}{2}$ and $v_1/v_2 = 2$; code is in `bandit_check/limit_theory.py` and `bandit_check/run_large_K.py`, which writes every number reported below to `bandit_check/results/large_K/summary.txt`. The finite-K values are the paper's exact ETC (integer τ, binomial w; $\tau \le 150$ for $K = 400$) and, for comparison, a learn-while-doing team that updates after every auction and re-paces its realised budget ("CE", computed by dynamic programming over periods, budget and posterior log-odds).

Limit objects and thresholds. The rate function is strictly concave in the belief only for $\rho < 0.035$ ($= \rho_c$ on the grid) and strictly convex in the belief for every $\rho \ge 1.375$ ($= \rho_v$ on the grid; checked up to $\rho = 10^4$); the best information structure starts to pay at $\rho = 0.305$ ($\Phi > 0$); both the best binomial experiment and perfect information start to pay at $\rho_1 = 0.4558$. Hence $\rho_c = 0.035 \le \rho_1 = 0.4558 \le \rho_v = 1.375$, as in Proposition 2(c). The entering branch is $\tau = \infty$, and on the grid $\Psi = \max\{\mathrm{VPI}, 0\}$ for every ρ: *the limit of the paper's ETC is the value of free perfect information itself.* Φ exceeds $\max\{\mathrm{VPI}, 0\}$ only on $(0.305, 0.8)$, by at most 0.0017 impressions per period (at $\rho \approx 0.455$). The asymptotics of Proposition 2 are confirmed: $\mathrm{VPI}(\rho)/\sqrt{\rho} = -0.1428$ at $\rho = 10^{-4}$ (theory: $-c_1 = -0.1469$) and $\rho^2\mathrm{VPI}(\rho) = 13.89$ at $\rho = 3000$ (theory: $\mathrm{Var}(\theta) = 14.06$). At entry, $\mathrm{VPI}'(\rho_1) = 0.0378$ and $g_\rho(\pi_0; \rho_1) = 0.2526$, so the limit marginal-value ratio jumps from 1 to 1.150 (the finite-K counterparts are 1.22 for the smoothed ETC and 1.17 for CE at $K = 100$).

Table 1: Per-period learning premium $(V_K - W_K)/K$ against the limit.

			ETC (exact), $K =$				CE (always learning), $K =$			
ρ	VPI	Ψ	50	100	200	400	50	100	200	400
0.20	−0.0113	0	0	0	0	0	−0.0224	−0.0175	−0.0147	−0.0133
0.60	0.0049	0.0049	0	0.0010	0.0021	0.0030	−0.0041	−0.0001	0.0022	0.0034
1.00	0.0143	0.0143	0.0090	0.0110	0.0123	0.0132	0.0066	0.0100	0.0119	0.0130
2.50	0.0252	0.0252	0.0151	0.0187	0.0212	0.0228	0.0195	0.0220	0.0234	0.0242
6.00	0.0228	0.0228	0.0068	0.0114	0.0153	0.0181	0.0190	0.0207	0.0216	0.0222
10.0	0.0174	0.0174	0.0025	0.0055	0.0090	0.0119	0.0146	0.0158	0.0165	0.0169

Rates. The total learning cost $K\,[\Psi(\rho) - (V_K - W_K)/K]$ of the ETC team grows like $\log K$, as in Proposition 1(a): at $\rho = 2.5$ it is 0.50, 0.65, 0.80, 0.95 impressions for $K = 50, 100, 200, 400$,

and 0.75, 1.19, 1.68, 2.20 at $\rho = 10$. The CE team's cost (relative to VPI) is much smaller and grows more slowly: 0.28, 0.32, 0.35, 0.39 at $\rho = 2.5$ and 0.14, 0.16, 0.18, 0.20 at $\rho = 10$. The optimal probing length grows roughly like $\log K$, so $\tau_K^*/K \to 0$ as in Proposition 1(c): at $\rho = 1.5$, $\tau_K^* = 7, 10, 13, 15$ and $\tau_K^*/K = 0.14, 0.10, 0.065, 0.037$. Near entry, the adaptive-stopping ETC exceeds Ψ and moves toward Φ, as allowed by Proposition 1(d): at $\rho = 0.5$ its premium is 0, 0.0011, 0.0019, 0.0023 against $\Psi = 0.0016$ and $\Phi = 0.0026$.

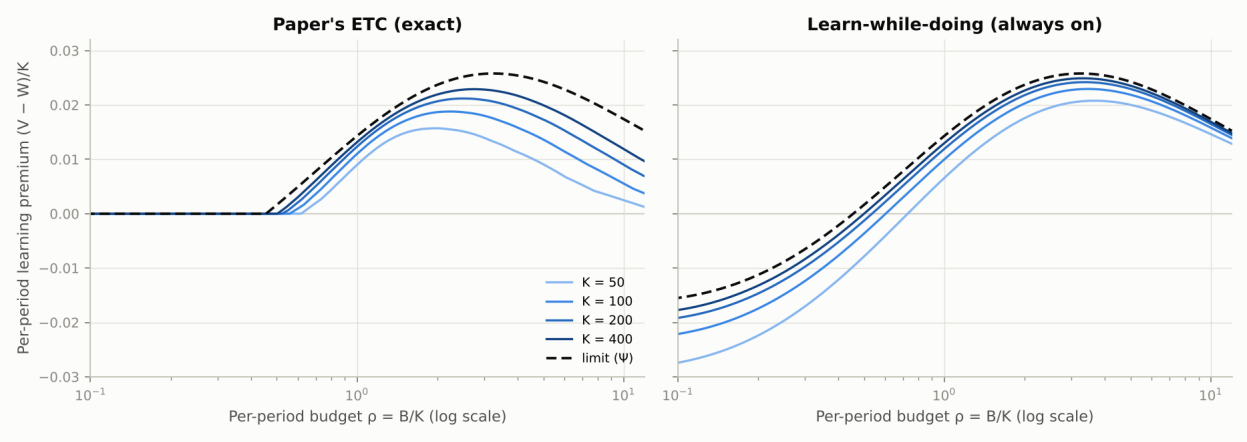


Figure 1: Per-period learning premium for $K = 50, 100, 200, 400$ (light to dark) and the limit (dashed). Left: the paper's ETC converges to $\Psi = \max\{\text{VPI}, 0\}$. Right: a learn-while-doing team converges to VPI, which is negative below ρ_1 (free information lowers the value of a budget-paced team).

Allocation. Table 2 reports the total budget per period $t = T/K$ at which learning first moves the split (toward the senior team) and at which the shift turns negative (toward the junior team). Both switch points converge to the same limit for the two protocols, as Corollary 1 predicts; at the limit switch $t = 1.5$ the junior team's benchmark budget per period is ≈ 0.48, just above ρ_1. The CE team's normalised shift converges quickly (e.g. $-0.0329, -0.0319, -0.0315, -0.0314$ at $t = 2.5$, limit -0.0311); the ETC team's converges more slowly, and its finite-K "re-concentration" at large t moves to ever larger budgets (Figure 2), consistent with Proposition 2(c).

Table 2: Switch points of the allocation shift $(a_K^L - a_K^E)/K$ in $t = T/K$ ($v_1/v_2 = 2$).

	$K = 50$	$K = 100$	$K = 200$	$K = 400$	limit
ETC: shift becomes positive	0.90	0.85	0.80	0.75	0.65
ETC: shift turns negative	1.95	1.85–1.95	1.75	1.65–1.70	1.50
CE: shift becomes positive	1.05	0.90	0.80	0.75	0.65
CE: shift turns negative	2.30	1.95	1.75	1.65	1.50

Open points. (i) Uniqueness of the zero of VPI and transversality at entry are verified numerically here and across the parameter sweep, but not proved. (ii) A proof of the CE limit. (iii) A probing design that implements the concave envelope (e.g. a planner-chosen probe bid with belief-dependent stopping) would lower the limit entry point from $\rho_1 = 0.456$ toward 0.305; Proposition 1(b) shows that no learn-then-commit protocol can do better than Φ.

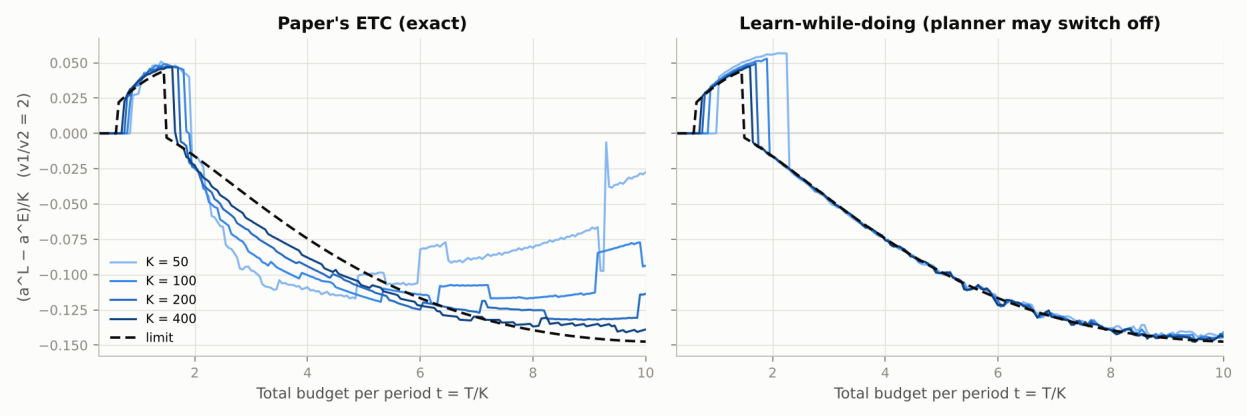


Figure 2: Normalised allocation shift $(a_K^L - a_K^E)/K$ for $K = 50, 100, 200, 400$ and the limit of Corollary 1 (dashed). Both protocols converge to the same limit.